

Right to Repair Calculator

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May 2025

What is the Right to Repair?

The right to repair is a movement advocating for consumers' ability to fix their own devices, such as smartphones, laptops, and agricultural equipment, without being forced to rely solely on manufacturers. This includes access to necessary tools, parts, and manuals. Supporters of the movement argue that enabling repairs promotes sustainability by reducing electronic waste, supports local economies through independent repair shops, and upholds consumer rights. Conversely, some manufacturers express concerns about intellectual property and safety, fearing that unauthorized repairs could compromise product integrity.

At its core, the right to repair is about restoring control to the consumer. Just as car owners expect to bring their vehicles to any mechanic of their choice, repair advocates believe electronic devices should offer similar freedoms. However, manufacturers sometimes design products to be intentionally difficult to service. For instance, some companies glue batteries in place, restrict software access, or withhold replacement parts from anyone outside of authorized service centers. These practices reduce consumer autonomy and make ownership more conditional than it ought to be.

Right to Repair in the U.S.

In recent years, numerous U.S. states have considered or enacted right-to-repair legislation. According to the National Conference of State Legislatures (NCSL), in 2023 alone, 30 states introduced bills aimed at requiring manufacturers to provide the necessary tools, software, parts, and manuals for consumers to perform their own repairs. Notably, Colorado passed a law ensuring that farmers can repair their own agricultural equipment, and Maine voters approved a measure granting independent auto shops and home mechanics access to the same diagnostic tools as dealerships. These legislative efforts reflect a growing recognition of the importance of repair rights in promoting consumer autonomy and environmental sustainability.

The fight for the right to repair is far from over. While some progress has been made, many states still have little to no legislation supporting repair access. In several cases, proposed bills have been delayed, watered down, or rejected entirely, often due to pressure from industry lobbyists and manufacturers. This ongoing resistance highlights the need for continued advocacy and public awareness to ensure that consumers nationwide gain the right to fix what they own.

The Ethics of the Right to Repair

Ownership includes the right to maintain and repair what one buys. When consumers cannot fix their devices or are forced to go through a single, manufacturer-approved channel, their freedom is restricted. This issue becomes especially pressing when manufacturers use restrictive repair policies to maximize profit at the consumer's expense.

Kantianism argues that humans should be treated as ends in themselves, never as means to another's gain. If a company limits repair options simply to increase revenue, it treats customers as a tool for financial gain rather than respecting their autonomy. Similarly, John Stuart Mill's utilitarianism evaluates actions based on whether they produce the greatest good for the greatest number (Mill). Allowing widespread repair access provides several clear benefits. It saves users money, encourages local repair businesses, reduces electronic waste, and fosters innovation by opening devices to deeper understanding. These outcomes outweigh any loss in new product sales. Ethics, from both Kantian and utilitarian viewpoints, strongly support the right to repair as a moral imperative. While it is reasonable for companies to worry about safety and liability, these concerns can be addressed with common-sense regulations that do not deny access altogether.

Our Project and My Contributions

Together with Samuel Goldsmith and Mina Boktor, I developed an ethical repairability calculator. Although tools like iFixit's repairability index already exist, we approached the issue from a different angle. Rather than focusing purely on technical complexity, we evaluated devices based on the ethical implications of their design. We asked whether the manufacturer

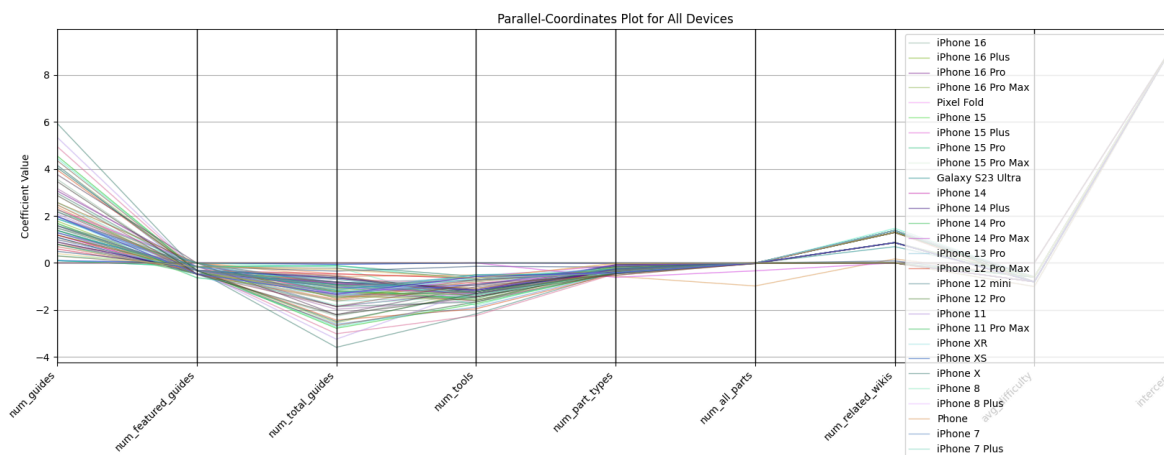
made reasonable efforts to support user repairs or whether the design choices created unnecessary obstacles.

To start, I manually re-evaluated a sample of devices listed on iFixit. I examined their reasoning behind each score, then reassigned scores that reflected ethical considerations, including documentation accessibility, tool requirements, and whether parts were intentionally restricted. Next, I used this adjusted dataset to train a linear regression model. The model incorporated several product features, including the number of repair guides and tools required, and used the Huber loss function to reduce the impact of outliers. The resulting formula produced a score that emphasized ethical design choices and penalized predatory practices.

Our formula:

$$\begin{aligned} & (9.3095 + (3.5655 * \text{num_guides}) - (0.4709 * \text{num_featured_guides}) \\ & - (2.2590 * \text{num_total_guides}) - (1.2294 * \text{num_tools}) - (0.0875 * \text{num_part_types}) \\ & - (0.001677 * \text{num_all_parts}) + (1.2970 * \text{rep_docs}) - (0.6026 * \text{avg_diff_num}) \end{aligned}$$

Here is a quick look at only some of the devices used to train our model:



Subset Device Contribution Graph

Relevance

This project has potential implications across several sectors. Everyday consumers benefit through reduced financial stress and greater independence when they can repair their own devices. Independent repair shops gain access to new opportunities, particularly as barriers fall

and more people seek affordable repair solutions. Policymakers can also use our findings to inform legislation that encourages ethical and accessible product design.

Of course, the project has its limitations. As a student initiative, we worked within a limited budget and timeframe. We could not afford to license data from paid APIs, which restricted the number of data points we could use. Additionally, we did not test repair difficulty firsthand for every device. Because of these constraints, the results cannot yet be generalized across all products or industries. However, the core methodology is adaptable and could be improved in future iterations.

To expand this work, we propose three next steps. First, we could partner with local repair shops to collect real-world data on repair time, success rates, and customer satisfaction. Second, we could conduct lifecycle assessments to measure environmental impacts. Third, we could use simulation models to examine how different repair policies affect consumers, businesses, and manufacturers.

Conclusion

The right to repair is about more than fixing a phone or replacing a battery. It is about ensuring people have the freedom to maintain what they own and to avoid being locked into corporate-controlled service networks. Our ethical repairability calculator redefines how we think about repair, offering a framework based on fairness and user empowerment. By combining machine learning with moral reasoning, we offer a new way to push the repair conversation forward. As more states consider legislation and as public awareness grows, tools like ours can help make repair access both practical and principled. Finally, by spreading the word about this pressing issue and informing people what devices and companies are actively working against their interests, we can force these companies to implement more ethical practices purely by making more well informed purchases

Works Cited:

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